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Anguita, J. V., Pozegic, T., Ahmad, M., & Silva, S. R. P. (2021). Layer-by-Layer Growth of Graphene Sheets over Selected Areas for Semiconductor Device Applications. ACS Applied Nano Materials, 4(5), 5211–5219. https://doi.org/10.1021/acsanm.1c00620
Document Version: Published (Version of record)

Published Version: https://doi.org/10.1021/acsanm.1c00620

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Layer-by-Layer Growth of Graphene Sheets over Selected Areas for Semiconductor Device Applications

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Cite This: ACS Appl. Nano Mater. 2021, 4, 5211−5219 Read Online

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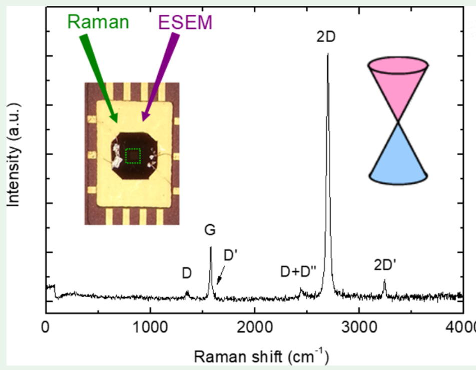
ABSTRACT: We report an *in situ* imaging method and use it to reveal the mechanism for the formation of extended size sheets of graphene (carpets) in few-layer graphene using the solid-state process, taking place via a layer-by-layer growth mechanism, which can result in a stack of separate individual layers of graphene. This mechanism is revealed by an imaging method that allows the use of conventional (unmodified) scanning electron microscopy to image graphene growth *in situ* and in real time. With this dynamic imaging, we reveal for the first time the dynamics of flake nucleation and growth and show the dynamics of flake coalescence to form extended size polycrystalline graphene carpets, allowing one to deduce a growth model. This growth method produces graphene flakes with Raman spectral characteristics that closely resemble those from exfoliated flakes obtained using the "Scotch-tape" method. The material is highly electronically intrinsic, with I_{2D}/I_G ratios as high as 5. The kinetics of electronic interconnectivity between flakes during blanket formation is imaged dynamically using a doping level contrast in an electron microscope in real time. Furthermore, the observations reveal that it is possible to maximize the time between the formation of each individual blanket, up to several minutes, thus facilitating the production of multiple decoupled graphene layers of precise thickness. This allows one to control the number of layers produced even when using catalysts of high activity and high-carbon solubility such as Fe.



KEYWORDS: *graphene in situ growth, real-time imaging, layer-by-layer growth, graphene sheets, selected-area, low-temperature-graphene growth, graphene microchip*

1. INTRODUCTION

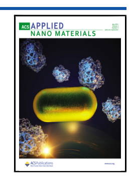
Graphene offers exceptional electrical, optical, and chemical characteristics. Among these are high carrier mobility and saturation velocity at room temperature,[1−3] broadband optical absorption,[4,5] selective permeability at a molecular level,[6,7] and fast transfer of protons.[8] It is believed that graphene will form the basis for the era of internet of things (IOT), where it will be used as a common platform not only making but also integrating a plethora of sensors, with electronics, high-frequency communications, and energy-scavenging components, to form one single IOT device.[9,11] However, this can only be realized, provided we develop a fast, reliable, and industrially viable method for the growth of high-quality graphene, which remains elusive even today .[10]

Part of the challenge stems from the strongly contrasting nature of the materials that are typically used for catalysis. On one extremum is Cu. This metal has negligible carbon solubility (~5 ppm at circa 900 °C),[12] and catalytic growth occurs by means of a surface-mediated mechanism. This provides an intrinsic self-limiting mechanism for controlling the growth to one single layer. Highly optimized processes have shown remarkably large-sized flakes, featuring dimensions in the centimeter scale and further extending to roll-to-roll productions.[12−14] However, the weak catalytic activity of Cu requires high growth temperatures, circa 1000 °C, and long growth times that extend over many hours, even for alloy-enhanced processes,[12−15] which are unfeasible for industrial applications.

In contrast, the superior catalytic activity of Fe and Ni permits significantly lower growth temperatures and shorter growth times concordant with manufacturing.[16] However, these feature high levels of carbon solubility at their operating temperature for catalysis, which typically leads to the formation of an uncertain number of layers. It is therefore required to develop a means for controlling the number of layers produced by such highly active catalysts. This would allow one to circumvent the high-temperature and extensive growth time requirements of Cu.

In addition to obtaining the optimal catalytic speed, it is also necessary that the growth process allows graphene flakes to grow in such a manner that allows them to coalesce together only

Received: March 2, 2021
Accepted: April 22, 2021
Published: May 5, 2021



around their perimeter, in order to form extended sheets of polycrystalline graphene. The extent of lattice disorder at the points of attachment between the flakes needs to be sufficiently small to allow electronic bands to bridge across flakes, enabling equal doping levels across the entire graphene sheet. It is therefore necessary to determine a suitable measurement technique that can ascertain the point during growth when this bridging occurs. We show here that this bridging point does not coincide with the point of physical contact of the flakes but occurs sometime after the physical contact.

We present a simple method whereby unmodified conventional scanning electron microscopy (SEM) can be used to image, analyze, and control the growth of graphene *in situ* and in real time. This imaging capability allows one to observe and control the aforementioned uncertainties associated with growth using Fe catalysis. In the past, highly sophisticated techniques for *in situ* imaging of graphene growth by catalytic chemical vapor deposition (cat-CVD) were developed by integrating an electron microscope into the cat-CVD chamber.[17−19] However, this involves a high level of technical complexity due to the need to inject the process gases (hydrocarbon and hydrogen) for graphene growth, while simultaneously imaging using a scanning electron microscope. This requires complex differential pumping stages and bespoke designs.[17−19] Here, we use the solid-state method,[20−22] where precise quantities of the carbon feedstock are provided as a thin solid layer of diamond-like carbon (DLC) instead of a gas. This layer is buried beneath the topmost Fe thin-film catalytic layer (see Methods). Upon thermal annealing (by means of a controlled electric current thought the Fe layer), the carbon diffuses along the grain boundaries to the top surface of the Fe layer, where it surface-migrates and forms graphene flakes via edge attachment. Without the need for process gases, it now becomes possible to use conventional SEM to image in high magnification the dynamics of graphene growing over the Fe surface. With this, we are now able to observe dynamically and *in situ* the effect of the temperature and time on graphene growth. In this paper, we examine graphene growth at different temperatures (700, 500, and 450 °C) as a function of time and observe the layer-by-layer growth of graphene at this elevated temperature in real time. We note that this imaging capability allows a user (in a manufacturing environment) to accurately determine the time of the growth process when the desired number of layers has been obtained and stop the heating (and growth), thus allowing one to produce the required number of layers for a particular application in a reproducible and reliable manner. This capability has been elusive to date.

With this imaging, we reveal for the first time the dynamics of all the stages involved in the growth of polycrystalline graphene layers over thin-film Fe: from nucleation and growth of hexagonal flakes to the dynamics of flake coalescence, consolidation into 2D layers, detachment from the substrate, and the growth of consecutive layers. We also use the Fermi level contrast in the scanning electron microscope to ascertain the electronic doping of individual flakes and to deduce the point of electronic coupling between adjacent flakes before and after physical flake coalescence. Our observations show that electronic coupling does not always coincide with the point of physical contact between flakes. This elucidates that the formation of additional chemical bonding or bond rearrangements at the interface between the flakes is necessary before electronic coupling is achieved.

The video imaging reveals the dynamics of layer formation: the detachment from the substrate and the formation of subsequent decoupled graphene layers. The videos reveal a mechanism that is based upon the layer-by-layer formation, where each layer is produced at different times and therefore independently from each other. Using this dynamic imaging, we are able to adjust the temperature profile in order to accentuate the time difference between the formation of consecutive layers by almost 5 min (258 s). This provides ample time to allow one to stop the growth process when the accurately controlled number of layers that are needed to produce has been obtained, as monitored in real time by SEM. This opens up the ability for real-time process control and reproducibility for fast-acting and low-temperature catalyst materials, which is a key requirement in design for manufacture. We postulate that this individual formation of each layer is the mechanism by which adjacent layers feature a random twist angle in their relative orientation and hence become electronically decoupled from each other.[5,23−25] This electronic detachment allows one to produce electronically intrinsic layers of high-quality, as ascertained from Raman spectroscopy.

2. EXPERIMENTAL SECTION

Industry-standard vacuum-deposition techniques were used to deposit all the thin-film materials required for growing graphene via the solid-state process inside the specimen chamber of the scanning electron microscope. A layer of DLC (50 nm) followed by the topmost Fe catalyst layer (200 nm) was vacuum-deposited on top of a thin silicon nitride membrane (50 nm) that is supported by a single-crystal silicon frame chip (Figure 1). Shadow masking was used to pattern the DLC

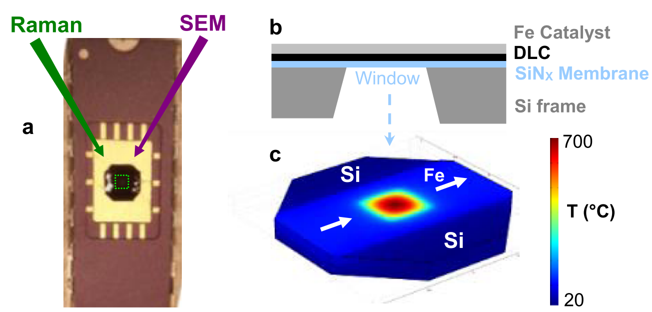


Figure 1. Chip design for *in situ* observation of graphene growth using the solid-state process and mounting on the chip carrier for analysis. (a) Growth imaged *in situ* by SEM and *ex situ* by Raman spectroscopy. (b) Cross-section schematic of the silicon frame and window membrane with all the relevant layers for catalytic growth using the solid-state method. The membrane dimensions have been exacerbated for display purposes. (c) Thermal model showing the heated window area and much cooler silicon frame areas, circa 20 °C. The Fe catalyst and the heater are the lighter blue region as labeled. The arrows indicate the current flow.

and Fe layers into a strip that covers the membrane, in order to funnel current through the window area for Joule heating, as shown in Figure 1. The heat capacity of the membrane is negligible compared to that of the silicon frame, and as a result, current injection through the Fe catalyst results in rapid thermal heating of the membrane, while the bulk of the silicon frame remains close to room temperature, as shown in Figure 1c. This arrangement provides strong localized heating on the selected areas suited to graphene growth, while temperature-sensitive electronics can remain at a lower temperature, below their degradation temperatures. Our thermal model also shows that the low-heat capacity of the membrane permits very high heating and cooling rates of the order of 10^3 K s^{-1}, which can be achieved by simple modulation of the Joule heating signal. The provision of such high heating and cooling

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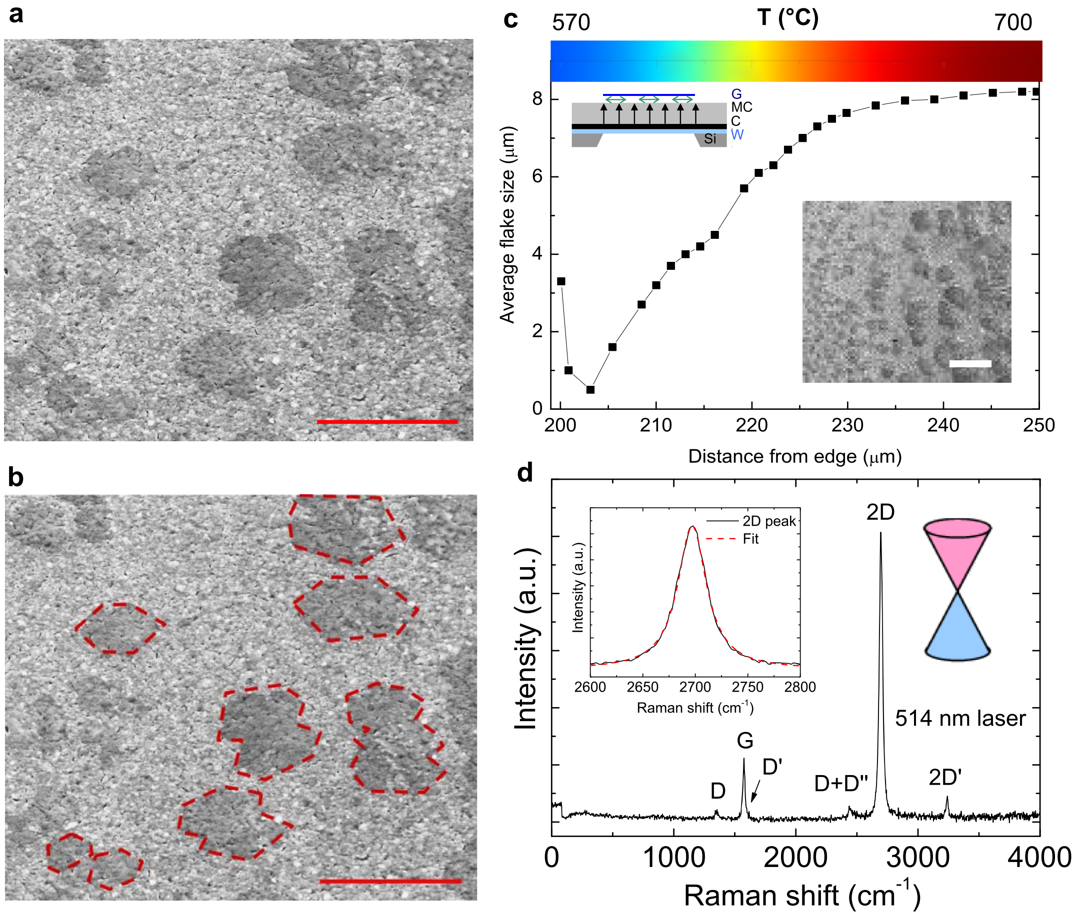

Figure 2. Growth of large and high-quality hexagonal graphene flakes on polycrystalline Fe during *in situ* observation by the solid-state process. The flakes are highly intrinsic and decoupled from the Fe substrate and feature a similar quality (I_D/I_G) as exfoliated graphene (a) Hexagonal flakes are many times larger than the grain size of the Fe catalyst, elucidating a departure from epitaxial needs during the growth stages (scale bar: 5 μm). (b) Highlights hexagonal flakes from (a), scale bar: 5 μm. (c) Size distribution of flakes as a result of the thermal gradient across the window. Regions of flake nucleation, growth, and coalescence are observed (inset bottom, scale bar: 10 μm). The growth mechanism based on diffusion across the metal catalyst (Fe) followed by surface migration (green) and flake growth (blue) is depicted (inset top). (d) Typical Raman spectrum from flakes showing a signature from high-quality single-layer intrinsic graphene, as evidenced from the low (I_D/I_G) and the single Lorentzian fitting on the 2D peak.

rates allows accurate control of the carbon diffusion through the Fe catalyst layer and quenching the growth of graphene once the required number of layers has been produced, and this is observed via the dynamic imaging. The chip was mounted on a chip holder and electrical connections are made, as shown in Figure 1a. Good thermal contact between the chip holder and the back of the Si frame ensured that the Si remains close to ambient temperature. The assembly was connected to a power supply via feedthroughs and placed in the specimen chamber of the scanning electron microscope.

Figure 2 shows the growth on the Fe catalyst after heating to 700 °C for a duration of 30 s. The temperature is deduced from the computer model. The heating and cooling rates were fast (~1 s) compared to the duration of the growth. The surface reveals the formation of hexagonal flakes, a characteristic of a Wulff construction that results from edge attachment-limited growth.[26,27] The flakes feature average dimensions of ~5 μm, which is significantly larger than the crystalline grain size of the Fe catalyst thin film of ~100−150 nm. This elucidates that the carbon can migrate significant distances along the Fe surface, even across grain boundaries, despite the stronger binding energy of carbon to Fe than to Cu, as shown in Figure 2c inset, and is able to reach and crystallize on the graphene flake, the mechanism of which is depicted in the inset of Figure 2c. The flakes are randomly oriented, which indicates either flake nucleation on single randomly oriented Fe grains, or the ability of the flakes to nucleate without epitaxial templating. It is observed that for the case of graphene growth on large grain size Cu foil, hexagonal flakes are favored only when operating under certain temperature conditions that allow the existence of noncrystalline Cu at the surface.[28−30] For example, by operating at high temperatures (~1000 °C) to produce a pre-melted Cu surface layer,[28] or even graphene growth on liquid Cu.[29,30] Subsequently, this suggests that the hexagonal flakes observed from our Fe films may result from catalytic/diffusion activity located at the amorphous regions in the grain boundaries, rather than activities in the crystalline grains of the Fe film catalyst.

Raman spectroscopy (performed ex situ) elucidates that the flakes exhibit a high level of crystalline quality, as shown in Figure 2d, revealing the same characteristic features as those obtained from high-quality exfoliated flakes that are free-floating (hence not coupled to any substrate).[31,32] The spectra exhibit a prominent G-peak, a weak D-peak (I_D/I_G peak ratio ~0.05), and a pronounced 2D band that fits a single Lorentzian curve, as shown in Figure 2d inset, which is a characteristic of single-layer graphene. The existence of a weak D-peak has been attributed to defects located at the edges of the flake[31] rather than crystalline defects within the bulk. The 2D′ peak (3500 cm^{-1}) is clearly visible and is observed in substrate-decoupled exfoliated graphene. A high I_{2D}/I_G ratio of ~4.5 elucidates that the flakes are highly electronically intrinsic. The strong resemblance of these Raman features with those from free-floating (i.e., substrate-free) exfoliated graphene flakes suggests that our graphene flakes are also fully decoupled from their substrate.[5,31,32] This reiterates the suitability of the solid-state process using the Fe catalyst for the growth of high-quality graphene.

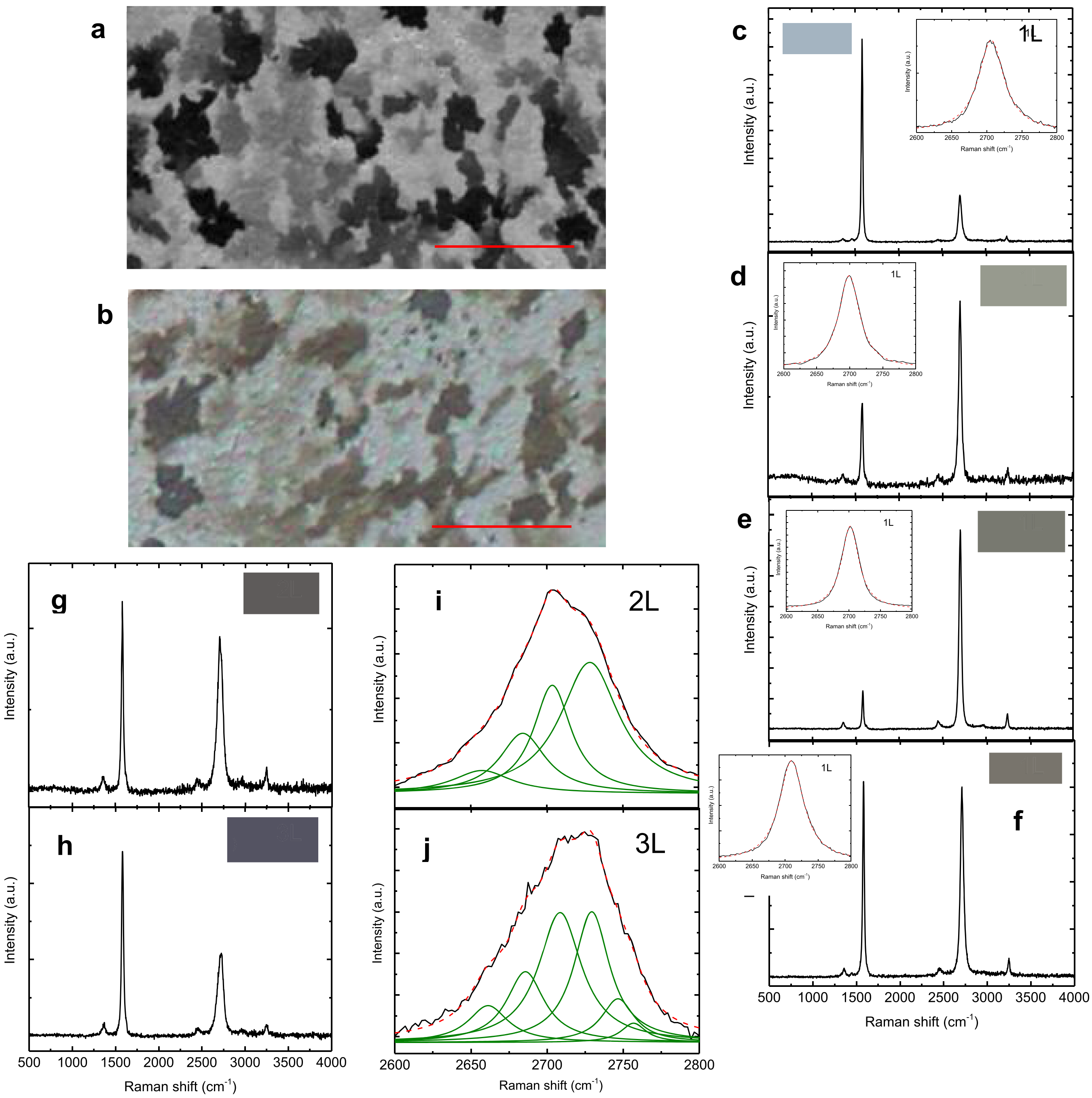

Figure 3. Graphene regions after extensive flake coalescence. These flakes are no longer hexagonal and feature different levels of contrast by (a) SEM secondary electron imaging and (b) optical microscopy. Note that (a,b) are of the same region, scale bar: 20 μm. (c−f) Raman spectra (*ex situ*) show variations according to the level of contrast shown in each inset. These indicate areas of single-layer graphene (lightest regions) and (g−h) few-layer graphene (darker regions). (i, j) Deconvolved 2D peaks from figures (g,h) to elucidate the presence of 2-layer (i) and 3-layer (j) graphene.

The size and nucleation rate of the graphene flakes as a function of distance on the membrane (which correlates to the process temperature, see temperature scale on Figure 2c) is shown in Figure 2c. The size distribution reveals that flake sizes increase with the temperature up to a critical size of ∼8 μm, until coalescence dominates; the flakes remain hexagonal across this temperature range. This suggests that the temperature values across this range can be used to control the rate of carbon diffusion across the Fe layer (thereby accounting for the flake-size distributions) but without affecting the flake morphology, which is controlled by parameters other than temperature within this range. This observation is consistent with the model for Cu,[28−30] where flake morphology is controlled predominantly by the crystalline structure of the catalyst rather than the temperature.

Increasing the temperature and duration of the growth allows the flakes to coalesce together, covering the full surface of the Fe catalyst film, as shown in Figure 3a,b. The electron microscopy image, as shown in Figure 3a, reveals areas of different contrasts, which correlate with the image from optical microscopy, as shown in Figure 3b. The Raman signal from the areas of equal contrast, as shown in Figure 3c−j with the contrast level shown in the inset of each figure, exhibit similar Raman signatures. The origin of the optical contrast is not well understood; however, graphene is readily observed on metal foils.[13]

It is observed that all levels of contrast, as shown in Figure 3, reveal the characteristic features of high-quality crystalline graphene.[31−33] These include sharp G peaks, low (<0.05) I_D/I_G peak intensity ratio values, and the presence of pronounced 2D and 2D′ peaks. For all cases,

https://doi.org/10.1021/acsanm.1c00620
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the intensity of the D-peak is significantly weaker than the G and the 2D peaks, indicating a high level of crystalline quality. The Raman spectra from the lighter regions, Figure 3c−f features a single-Lorentzian 2D peak (fittings in the figure inset), indicating that the regions are composed of either single-layer graphene or multiple layers of electronically decoupled graphene. The lightest regions, as shown in Figure 3c, exhibit I_{2D}/I_G ratio values of ∼0.2. These elucidate graphene regions that are nonintrinsic, possibly as a result of strong electronic coupling to the Fe substrate. This layer is sometimes referred to as the "zero" layer as it remains strongly coupled (and bonded) to the substrate. The light-colored appearance in optical microscopy results from a lack of direct electronic transitions that are available for the case of partially filled bands. These transitions become more available as the graphene becomes more intrinsic, as shown in Figure 3d, $I_{2D}/I_G \sim 2$, and eventually fully intrinsic, as shown in Figure 3e, $I_{2D}/I_G \sim 4.5$, which is observed as a gradual darkening of graphene. The changing level of doping is also observed by the changing contrast in SEM.[34] The darkest regions in the figure, as shown in Figure 3g,i, consist of multiple layers that are coupled to each other, yet free from the substrate. This is ascertained from the four and six Lorentzian curves that can be fitted into their 2D peaks, as shown in Figure 3h,j, indicating the presence of multiple layers for the darker regions.

Reducing the growth temperature to 500 °C has the effect of reducing the rate of carbon diffusion, thereby reducing the rate of growth, as shown in Figure 4 and Video S1. The figure reveals a featureless Fe surface at the point of heating, as shown in Figure 4a. No significant change in the appearance of the surface is observed throughout the first ∼200 s from the initial heating (deduced from the model). After this time, a distribution of flakes nucleating on the Fe surface are observed, as shown in Figure 4b. For the subsequent ∼500 s, the flakes are observed to be growing in size and coalesce to form the first graphene layer, as shown in Figure 4c−f. The first graphene layer forms completely after ∼680 s, as shown in Figure 4f. Interestingly, it is observed that no further change is observed on the surface for a duration of 258 s (until a total process time of 938 s, Figure 4g), when flakes from the second layer begin to nucleate and appear in the image, subsequent to layer one deposition.

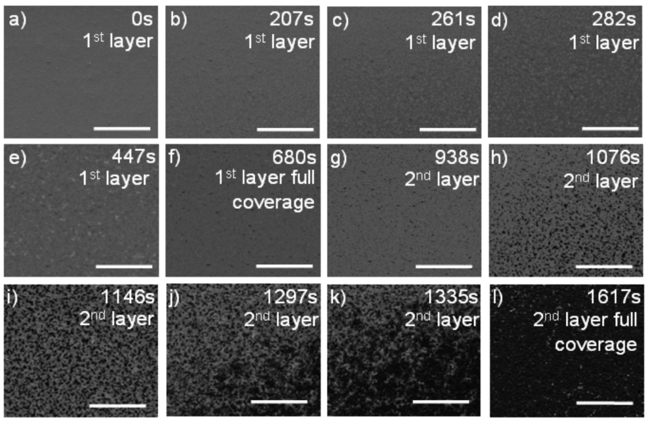

Figure 4. Direct imaging of layer-by-layer growth of graphene extensively over a large area (in comparison with the flake and Fe grain size), using the solid-state method (all scale bars: 20 μm). A video of this growth in real time is available (S1). The process has been adjusted to elongate the delay between the end of the first layer and beginning of the second layer (up to 258 s) enabling ample time for end-point detection and halt the process after the first layer. (a−f) Nucleation, growth, and coalescence of flakes, producing a continuous single layer of graphene, over this extended area. This single layer is stable for a period of 258 s (f−g) before the second layer begins to grow. (g−l) Growth of the second layer, reaching full coverage.

3. DISCUSSION

It is remarkable that no growth takes place on the surface between the formation of the first layer at 680 s and appearance of flakes from the second layer (Figure 4g), at 938 s. This allows for a process window greater than 4 min (258 s) where a uniform single sheet of graphene exists without the formation of discontinuous regions from subsequent layers. Once this time is exceeded, the flakes from the second layer begin to nucleate, as shown in Figure 4g, and grow until the second layer is complete, as shown in Figure 4h−l. The Raman spectrum shows that the 2D peak can be fitted to a single Lorentzian curve, indicating electronic decoupling between the layers.[31−34] This decoupling is reported to occur as a result of the formation of each layer at different times and independently from each other.[5,31−34] This also introduces a random twist angle between the orientation of the layers.[5]

The area coverage as a function of time, as shown in Figure 5a, shows that for the first layer, the rate of coverage reduces with time. This observation is remarkably similar to that observed for growth using the vapor method. However, the origin for this decrease is different for both cases. For the vapor growth, the decrease originates from the growth of graphene itself, which hinders the exposure of the hydrocarbon vapor to the catalytic surface. This effect, however, does not occur for the solid-state method. The origin for the reduced diffusion rate for the solid-state method is not known but is likely to be caused by a reduction in the amount of carbon available at the source as time progresses and also, an effective thickening of the catalytic layer due to the incorporation of carbon, which has the effect of increasing the diffusion path, thereby increasing diffusion times. The Avrami plot, as shown in Figure 5b, shows linearity, which indicates this effect. The linearity indicates a Johnson−Mehl−Avrami−Kolmogorov (JMAK) growth model, which elucidates the mechanism for the formation of the extended single, polycrystalline graphene layer.

From the observations, it is possible to determine a Fickian diffusion model, as shown in Figure 5, that predicts the time taken for sufficient carbon atoms to appear on the surface to produce one complete graphene layer, for growth conditions at different temperatures, as shown in Figure 5c,d. Figure 5d shows that this growth method allows adjusting the time taken for full-layer formation by several orders of magnitude, depending on the growth temperature. This way, it is possible to obtain layer-by-layer growth time spacings in the order of hundreds of seconds, by selecting a growth temperature around 600 °C. This time spacing amply allows for process control, for selecting the number of layers that is required for a specific application. It also allows to produce each layer in a time frame that is compatible with mass production (hundreds of seconds), which contrasts with the many hours for the case of copper catalysis. The model also predicts full layer formation timescales in the order of seconds for temperatures circa 800 °C, which typically results in the observed multilayer growth for this type of catalysis. At the opposite temperature end, the model predicts growth times of several hours at the extreme low temperatures (below 400 °C). However, it is likely that this temperature does not provide sufficient thermal energy to activate the dissociation processes that are required for the formation of graphene.

The dynamics of the flake coalescence process and electronic coupling to their nearest neighbors was also observed by imaging in real time at a higher magnification. For this, it was necessary to reduce the speed of the growth further by reducing the growth

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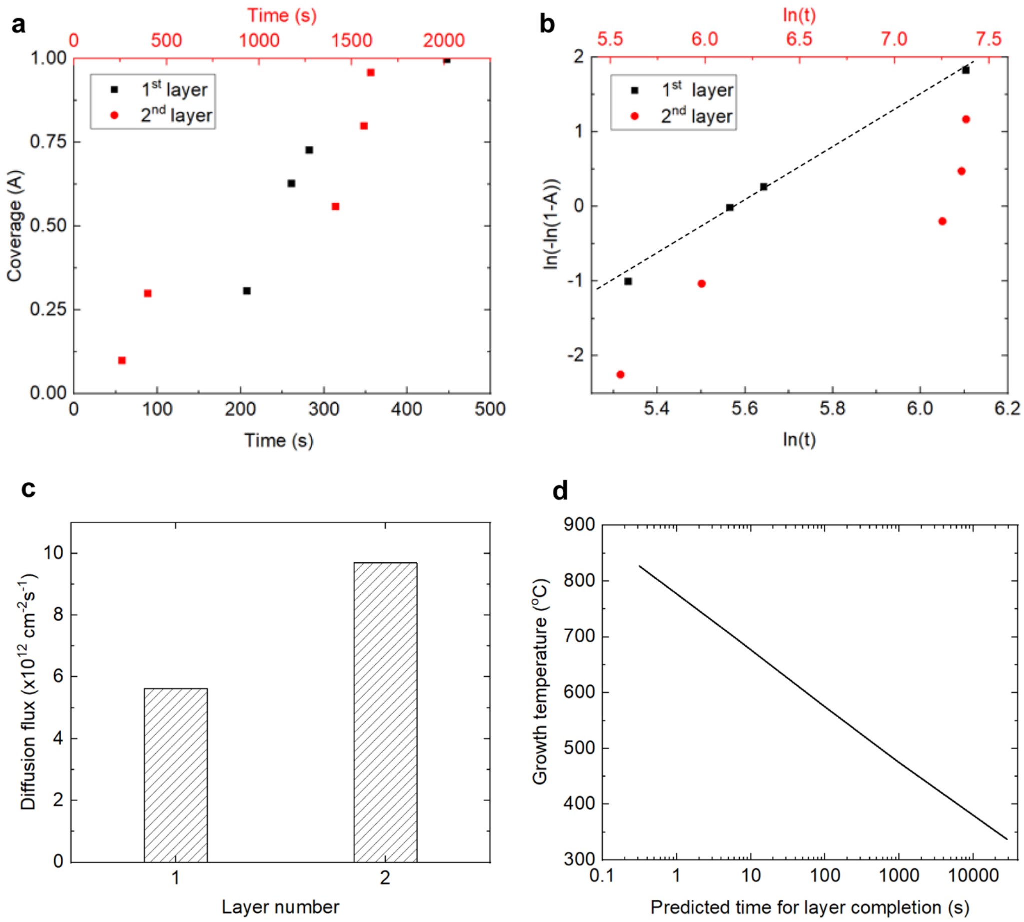

Figure 5. Kinetics of the growth and predictive model showing the (a) area coverage as a function of time for the first and second layers, (b) Avrami plot indicating linearity for the first layer, evidencing a Johnson−Mehl−Avrami−Kolmogorov (JMAK) growth model, and deviation from this model for the second layer, (c) diffusion flux of carbon for the two layers, and (d) predictive model (diffusive) which predicts the time required for layer completion as a function of temperature, based on diffusion data.

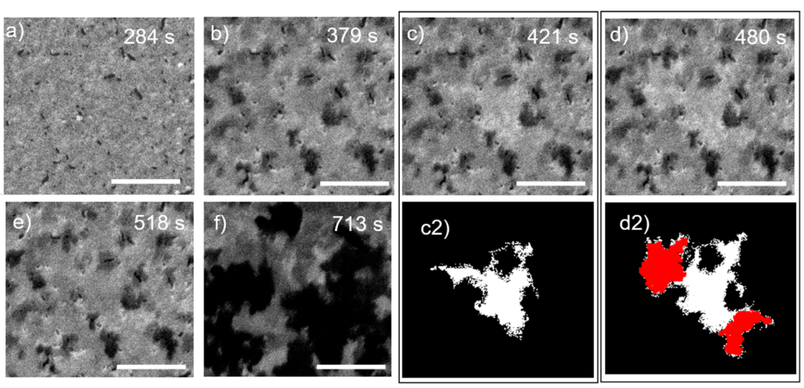

Figure 6. Imaging of flake coalescence sequence and electronic coupling to neighboring flakes, via monitoring the contrast of individual flakes of the graphene surface, scale bars: 10 μm. (a) Fe surface exhibiting small-grain-size Fe. Dark patches reveal preferential sites for carbon diffusion to the surface. (b) Graphene coverage of the surface by surface migration. (c) The contrast of a particular flake quickly becomes lighter compared to the background, as a result of becoming differently electronically doped than the rest of the blanket. (d) The flake becomes connected to two neighboring flakes, highlighted in red in (d2). (e) The extended flake quickly regains the same contrast as that of the graphene blanket, possibly as a result of becoming electronically coupled to it. (f) Growth of the second graphene layer on top of the first layer. (c2,d2) Contrast-enhanced images of (c,d), respectively, to highlight the shape of the flake being discussed.

temperature to an estimated 450 °C, as determined by the computer model. The process is captured in Figure 6a−f and Video S2. The images and videos show flakes within the graphene blanket experiencing rapid changes in their contrast but not in their morphology. For example, Figures 6b−e show a particular region within the graphene blanket, where its contrast changes quickly from dark, as shown in Figure 6b, to light, as shown in Figure 6c, and back to dark again, as shown in Figure 6e. The region where this contrast change occurs is highlighted in Figure 6c(2). A further two adjacent regions exhibited similar

https://doi.org/10.1021/acsanm.1c00620
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behavior, as shown in Figure 6d, where these are highlighted in red in Figure 6d(2). After some 38 s, the whole region regained the same contrast in the SEM as the rest of the blanket, as shown in Figure 6e, which suggests the point of electrical coupling between all the flakes within the blanket.

The flake imaged in Figure 6c is nonhexagonal, which suggests that it has grown and has merged with the graphene blanket and thereby is in physical contact with the blanket. The uniform SEM contrast within this flake, as shown in Figure 6c,d, but different from the rest of the blanket suggests a uniform doping level within the flake but different from that of the blanket in its surroundings, despite it being in physical contact. The rapidly changing contrast level of this flake and those in its vicinity, as shown in Figure 6d, suggest a contrast mechanism based on electronic doping rather than the contrast from structural changes, as reported by other groups.[34] For this type of contrast mechanism, the n-type graphene appears darker than intrinsic or p-type graphene. We propose complementing the analysis described here with a suitable SEM tool that incorporates a facility to perform micro-Raman spectroscopy *in situ*, in order to arrive at a dedicated facility for industrial implementation.

In order to demonstrate the role of C surface migration in obtaining graphene of high-quality (and thereby determine the growth mechanism), we have attempted to grow graphene in a way that blocks this surface migration and compared it with the normally grown one. Blocking surface migration was achieved by reversing the geometry of the solid-state process and growing the graphene along the interface between the Fe and nitride windows, as shown in the inset of Figure 7. For this, we located

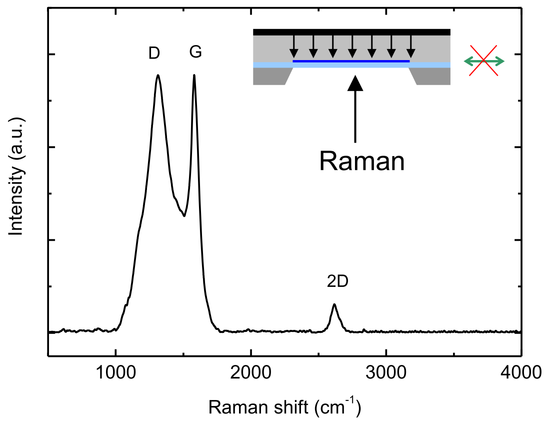

Figure 7. Raman spectra from the reverse orientation of the solid-state growth mechanism to monitor the carbon diffusion through the Fe to the Si_3N_4/Fe interface (inset). The figure indicates the formation of amorphous carbon instead of graphene, as a result of the lack of lateral migration along the interface (inset right).

the Fe catalyst layer directly above the Si_3N_4 membrane and placed the DLC on top of Fe. With this arrangement, carbon diffusion from the DLC across Fe, leads to the Fe/Si_3N_4 interface, where it is not able to surface migrate. Raman spectroscopy, performed through the nitride window, showed broad D and G bands of similar intensity, and a weak and broad 2D band (Figure 7), indicative of amorphous carbon. This is despite the carbon having diffused through the Fe catalyst layer. This result highlights the necessity for lateral surface migration and further evidences graphene formation via edge-additive growth. It also suggests that the Fe does not play a role in crystallographic arrangements of the C atoms, which further elucidate reports of independence between the catalyst crystal orientation and the resulting flake orientation.[29,30,35] This result suggests that impeding surface diffusion results in a significant reduction of the ability for carbon to form graphene of high-quality.

4. CONCLUSIONS

We have demonstrated a simple imaging technique for *in situ* imaging the growth of graphene in real time, using conventional SEM and Fe catalysis with the solid-state process. The images reveal the layer-by-layer growth of high-quality graphene layers on industrially relevant substrates. We have shown the suitability of Fe for catalysis using the solid-state method, which has produced graphene quality similar to that of free-floating exfoliated graphene. We have also proven the ability to control the number of layers with our real-time imaging arrangement for growth.

5. METHODS

Thin-film growth was performed using a sputtering system from JLS, using a 4 in. diameter target, argon gas with a flow rate of 20 sccm, 5 mTorr pressure, 280 V bias, and 0.2 A current. The growth of DLC was made in a 1000 N system from Surrey NanoSystems, using 50 sccm H_2 and 10 sccm C_2H_2 at a combined pressure of 200 mTor. The plasma power was 200 Watts. The dynamic imaging of the growth was made using an ESEM, FEI quanta 200 FEG operated in the standard SEM mode, using an acceleration voltage of 10 kV and an in-column detector for secondary electrons. Raman measurements were made using a system from Renishaw.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsanm.1c00620.

Direct imaging of layer-by-layer growth of graphene extensively over a large area using the solid-state method, scale bar: 20 μm (MP4)

Imaging of flake coalescence sequence and electronic coupling to neighboring flakes, via monitoring the contrast of individual flakes of the graphene surface, scale bar: 10 μm (MP4)

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Complete contact information is available at: https://pubs.acs.org/10.1021/acsanm.1c00620

https://doi.org/10.1021/acsanm.1c00620
ACS Appl. Nano Mater. 2021, 4, 5211−5219

Author Contributions

The manuscript was written through contributions of all the authors. J.V.A and S.R.P.S. designed the experiments, did the growth, and wrote the paper. T.R.P. helped with the Raman measurements, M.A. contributed to the manuscript. All the authors have given approval to the final version of the manuscript

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We acknowledge the EPRSC grant number EP/L02263X/1.

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https://doi.org/10.1021/acsanm.1c00620
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